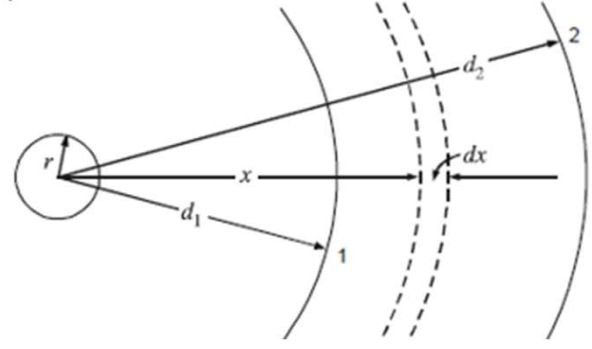
Methodology for Deriving External Flux Linkage

- Consider a thin cylindrical shell of radius x and thickness dx outside the conductor.
- Apply Ampere's Circuital Law to determine the magnetic field intensity (H).
- Calculate the magnetic flux density using: B = μH
- Determine the elemental flux (dΦ) through the shell.



- Since external flux links the entire conductor current, the elemental flux linkage is equal to the elemental flux.
- Integrate dψ between the specified radial distances (d1 to d2) to obtain the total external flux linkage.